DSTA1303 - Statistical Distributions

Bernouli and Binomial Distributions

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Bernoulli Distribution

- The **Bernoulli distribution** is the simplest named probability distribution.
- It serves as a **building block** for the Binomial, Geometric, and Negative Binomial distributions.
- We use the term *"success"* to mean *"the thing we're looking for"*, regardless of whether it is good or bad.
- The notation for a Bernoulli distribution is:

$$X \sim \text{Bernoulli}(p)$$

  where p is the probability of success, and the failure probability is $q = 1 - p$.

Bernoulli Random Variables

- The common thread: **One success or one failure**?
- Example scenarios:
  - Asking one person a yes/no question (yes = success, no = failure)
  - Flipping a coin (heads vs. tails)
  - Checking if a part is defective (yes or no)
  - Classifying a pet's size (large or small)
- **Key idea:** One trial, which results in either a success or a failure
- The random variable X:
  - $X = 0$ if the outcome is a failure
  - $X = 1$ if the outcome is a success
- The parameters:
  - p = probability of success
  - $q = 1 - p$ = probability of failure

Bernoulli: PMF, Expectation, and Variance

Probability Mass Function (PMF):

$$P(X = 1) = p$$
$$P(X = 0) = q = 1 - p$$

Expected Value:

$$\mathbb{E}(X) = p$$

Variance:

$$\mathrm{Var}(X) = pq = p(1 - p)$$

Example: Peanut Butter First?

Scenario: 95% of people put peanut butter on first when making a peanut butter and jelly sandwich. A person is selected at random.

(a) Why is this a Bernoulli distribution situation?
Single trial with a success/failure outcome. Parameter: $p = 0.95$

(b) What do we consider a "success"?
Finding a person who puts peanut butter on first.

(c) What is the probability the person puts jelly on first?
$q = 1 - p = 0.05$

(d) Expected number of people who put peanut butter on first (when asking one person):
$\mathbb{E}(X) = p = 0.95$

(e) Standard deviation:

$$\text{Var}(X) = pq = (0.95)(0.05) = 0.0475$$

$$\sigma_X = \sqrt{pq} = \sqrt{0.0475} = 0.2179$$

Remark: Expected Value and Variance

- If X is a Bernoulli random variable, then:

$$\mathbb{E}(X) = (0)(q) + (1)(p) = p$$

- The second moment:

$$\mathbb{E}(X^2) = (0^2)(q) + (1^2)(p) = p$$

- Variance formula:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}(X))^2] = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$$

$$\mathrm{Var}(X) = p - p^2 = p(1 - p) = pq$$

Example: Expected Gain in a Die Game

Scenario: A player needs to roll a 5 or 6 to win (probability $= \frac{2}{6} = \frac{1}{3}$). If she wins, she earns \$15; if she loses, she forfeits \$9.

Let $X \sim \text{Bernoulli}\left(\frac{1}{3}\right)$ where:

- $X = 1$: player wins, gain $= \$15$
- $X = 0$: player loses, loss $= -\$9$

Expected winnings:

$$\mathbb{E}(\text{winnings}) = (-9)\left(\frac{2}{3}\right) + 15\left(\frac{1}{3}\right) = -6 + 5 = -1$$

Formal notation:

$$\mathbb{E}[f(X)] = f(0)P(X = 0) + f(1)P(X = 1) = (-9)\left(\frac{2}{3}\right) + 15\left(\frac{1}{3}\right) = -1$$

Conclusion: The player expects to **lose \$1** at this point in the game.

Binomial Random Variables

- A **Binomial random variable** counts the number of successes in n independent trials.
- It is the sum of n independent Bernoulli random variables with the same success probability p:

$$X = X_1 + X_2 + \cdots + X_n, \quad \text{where each } X_i \sim \text{Bernoulli}(p)$$

- If $n = 1$, then $\text{Binomial}(1, p) \equiv \text{Bernoulli}(p)$.
- For a Binomial random variable, both n (number of trials) and p (success probability) must be known in advance.

Binomial Distribution Notation

Example: Suppose $n = 8$ trials, with X_1, X_2, X_5 and X_7 indicating success. Let:

$$X_1 = X_2 = X_5 = X_7 = 1, \quad \text{others} = 0$$

$$X = X_1 + \cdots + X_8 = 1 + 1 + 0 + 0 + 1 + 0 + 1 + 0 = 4$$

Then $X \sim \text{Binomial}(8, p)$ and the total number of successes is 4.

- The notation for a Binomial random variable is:

$$X \sim \text{Binomial}(n, p)$$

- Where:
  - n: total number of trials
  - p: probability of success on each trial

Remark: Binomial Coefficients

- The number of ways to have j successes in n trials (for $0 \leq j \leq n$) is given by the **binomial coefficient**:

$$\binom{n}{j} = \frac{n(n-1)(n-2)\cdots(n-j+1)}{j!} = \frac{n!}{j!(n-j)!}$$

where

$$n! = n \times (n-1) \times \cdots \times 1, \quad j! = j \times (j-1) \times \cdots \times 1$$

and by definition $0! = 1$.

- **Example:** For $n = 8$, the values are:

$$\binom{8}{0} = 1, \quad \binom{8}{1} = 8, \quad \binom{8}{2} = 28, \quad \binom{8}{3} = 56,$$
$$\binom{8}{4} = 70, \quad \binom{8}{5} = 56, \quad \binom{8}{6} = 28, \quad \binom{8}{7} = 8, \quad \binom{8}{8} = 1.$$

Binomial Random Variables

- The common thread: Number of successes in a fixed number of independent trials with the same probability of success on each trial.
- Examples:
  - Asking 10 people a yes/no question each.
  - Number of 5's in 100 die rolls.
  - Number of defective items in a sample of 12 from an assembly line.
  - Number of seniors in a survey of 8500 students.
- Key features:
  - Fixed number of independent trials.
  - Same success probability p for each trial.
- Random variable:

$$X = \text{number of successes in } n \text{ trials}, \quad 0 \leq X \leq n$$

- Parameters:

$$n = \text{total number of trials}, \quad p = \text{probability of success}, \quad q = 1 - p$$

- If $n = 1$, the Binomial reduces to a Bernoulli.

Binomial Distribution: PMF, Expectation, and Variance

Probability Mass Function (PMF):

$$P(X = x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, \ldots, n$$

Expected value:

$$\mathbb{E}(X) = np$$

Variance:

$$\mathrm{Var}(X) = npq$$

Explanation of the PMF:

- $\binom{n}{x}$: number of ways to arrange x successes in n trials
- p^x: probability of x successes
- q^{n-x}: probability of $n - x$ failures

Example: Snowy New Year's Days

Scenario: Predict snow on New Year's Day over 31 years. Probability of snow each year: 0.3. Trials assumed independent.

(a) Number of trials: $n = 31$

(b) X = number of snowy New Year's Days in 31 years, $X = 0, 1, \ldots, 31$

(c) Why Binomial? Fixed number of independent trials, same success probability $p = 0.3$, counting successes.

(d) Probability of exactly 5 snowy days:

$$P(X = 5) = \binom{31}{5}(0.3)^5(0.7)^{26} = 0.03876$$

(e) Probability of at least 1 snowy day:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \binom{31}{0}(0.3)^0(0.7)^{31} = 1 - 0.000016 = 0.999984$$

Expected Value and Variance of a Binomial Random Variable

- Let $X_1, \dots, X_n$ be independent Bernoulli random variables with success probability p.
- Then $X = X_1 + \cdots + X_n \sim \text{Binomial}(n, p)$.
- Using linearity of expectation:

$$\mathbb{E}(X) = \mathbb{E}(X_1 + \cdots + X_n) = \mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n) = np$$

- Since the X_j are independent, variances add:

$$\mathrm{Var}(X) = \mathrm{Var}(X_1 + \cdots + X_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n) = npq$$

where $q = 1 - p$.

Example: Counting 4's and 5's in Die Rolls

- Roll a die n times. Let X = total number of 4's and 5's.
- Success = rolling a 4 or 5, so $p = \frac{2}{6} = \frac{1}{3}$.
- The PMF of X is:

$$p_X(j) = \binom{n}{j} \left(\frac{1}{3}\right)^j \left(\frac{2}{3}\right)^{n-j}, \quad j = 0, 1, \ldots, n$$

  and $p_X(x) = 0$ otherwise.
- Expected value:

$$\mathbb{E}(X) = np = \frac{n}{3}$$

- Variance:

$$\mathrm{Var}(X) = npq = n \times \frac{1}{3} \times \frac{2}{3} = \frac{2n}{9}$$

Example: Majority Putting Peanut Butter First

Suppose 95% of people put peanut butter on first. Five people are asked independently.

Let $X \sim \text{Binomial}(5, 0.95)$ be the number who put peanut butter on first.

$$P(X = j) = \binom{5}{j}(0.95)^j(0.05)^{5-j} = \binom{5}{j}\frac{19^j}{20^j}\frac{1^{5-j}}{20^{5-j}} = \binom{5}{j}\frac{19^j}{20^5}$$

for $j = 0, 1, 2, 3, 4, 5$, and zero otherwise.

Majority means $X = 3, 4,$ or 5:

$$\begin{aligned} P(X \geq 3) &= P(X = 3) + P(X = 4) + P(X = 5) \\ &= \binom{5}{3}\frac{19^3}{20^5} + \binom{5}{4}\frac{19^4}{20^5} + \binom{5}{5}\frac{19^5}{20^5} \\ &= 0.0214 + 0.2036 + 0.7738 = 0.9988 \end{aligned}$$

Remark: Sums of Binomial Random Variables

- Consider independent Binomial random variables $X_1, X_2, \ldots, X_m$ with the same success probability p, and number of trials $n_1, n_2, \ldots, n_m$.
- The sum

$$Y = X_1 + X_2 + \cdots + X_m$$

counts the total successes in

$$N = n_1 + n_2 + \cdots + n_m$$

independent trials.
- Therefore, Y is also Binomial with parameters N and p:

$$Y \sim \text{Binomial}(N, p)$$

- The probability mass function is:

$$p_Y(j) = \binom{N}{j} p^j q^{N-j}, \quad 0 \leq j \leq N$$

and zero otherwise.

Example: Total Number of Rap Songs in Combined Playlists

- Dominique, Raymond, and Samantha have playlists with 30, 70, and 90 songs respectively.
- Each song is rap with probability $p = 0.40$.
- Let X_1, X_2, X_3 be Binomial random variables representing the number of rap songs:

$$X_1 \sim \text{Binomial}(30, 0.40), \quad X_2 \sim \text{Binomial}(70, 0.40), \quad X_3 \sim \text{Binomial}(90, 0.40)$$

- The total number of rap songs:

$$Y = X_1 + X_2 + X_3$$

- By the sum property of Binomial variables:

$$Y \sim \text{Binomial}(190, 0.40)$$

where $N = 30 + 70 + 90 = 190$.